

CSCA 数学/Math 15th March 2026

1. Let $A = \{x \mid -2 < x \leq 3\}$ be a set. Which of the following statements is correct? ()
 - (a) $2 \notin A$
 - (b) $-3 \in A$
 - (c) $3 \in A$
 - (d) $-2 \in A$
2. Let $M = \{x \mid -5 \leq x < 2\}$, $N = \{x \mid -4 \leq x < 2\}$ be two sets. Then $M \cap N =$ ().
 - (a) $(-4, 2]$
 - (b) $(-5, 2)$
 - (c) $[-4, 2)$
 - (d) $[-5, 2]$
3. The solution set of the inequality $x^2 - x - 6 > 0$ is ().
 - (a) $\{x \mid -2 < x < 3\}$
 - (b) $\{x \mid x < -2 \text{ or } x > 3\}$
 - (c) $\{x \mid x < -2\}$
 - (d) $\{x \mid x > 3\}$
4. The domain of the function $f(x) = \frac{1-x}{1+x}$ is ().
 - (a) $(-\infty, -1) \cup (-1, +\infty)$
 - (b) $(-\infty, 0) \cup (0, +\infty)$
 - (c) $\mathbb{R}$
 - (d) $(-\infty, 1) \cup (1, +\infty)$
5. If $a = -2$, $ab < 0$, then the point (a, b) belongs to ().
 - (a) the second quadrant
 - (b) the fourth quadrant
 - (c) the first quadrant
 - (d) the third quadrant

6. Let $\alpha = \frac{5}{6}\pi$. Then $\cos \alpha = ()$.

(a) $-\frac{1}{2}$

(b) $\frac{\sqrt{3}}{2}$

(c) $\frac{1}{2}$

(d) $-\frac{\sqrt{3}}{2}$

7. Over its domain, the function $f(x) = \ln(x^2)$ is ().

(a) odd

(b) uncertain about its parity

(c) neither odd nor even

(d) even

8. The inverse function of $y = \frac{3x - 2}{2x + 1}$ is ().

(a) $y = \frac{2x - 3}{x + 2}$

(b) $y = -\frac{x + 2}{2x - 3}$

(c) $y = -\frac{2x - 3}{x + 2}$

(d) $y = \frac{x + 2}{2x - 3}$

9. The function $f(x) = \sin x$ is monotonic increasing over ().

(a) $[-\pi, \pi]$

(b) $[0, \pi]$

(c) $(-\infty, +\infty)$

(d) $[-\frac{\pi}{2}, \frac{\pi}{2}]$

10. Let $\{a_n\}$ be an arithmetic sequence, with the first term $a_1 = 2$ and common difference $d = 3$. Then $a_{100} = ()$.

(a) 298

(b) 302

(c) 305

(d) 299

11. The solution set of the inequality $\frac{2x-3}{3x-4} < 0$ is ().

(a) $[\frac{4}{3}, \frac{3}{2}]$

(b) $(\frac{4}{3}, \frac{3}{2})$

(a) $(\frac{4}{3}, \frac{3}{2}]$

(a) $(\frac{4}{3}, \frac{3}{2}]$

12. The geometric mean of 3 and 12 is ().

(a) ± 6

(b) ± 4

(c) ± 10

(d) ± 8

13. The angle of inclination of the line $x + \sqrt{3}y - 2 = 0$ is ().

(a) 60°

(b) 150°

(c) 120°

(d) 30°

14. If a triangle $\triangle ABC$ has sides $BC = 5$, $AC = 12$, $AB = 13$, then $\sin A$ is ().

(a) $\frac{5}{13}$

(b) $\frac{5}{12}$

(c) $\frac{7}{12}$

(d) $\frac{12}{13}$

15. In a geometric sequence $\{a_n\}$, if $a_1 = 4$ and $a_4 = 32$, then a_3 is ().

(a) 8

(b) -8

(c) 16

(d) -1

16. Which of the following points is in the third quadrant? ().
- (a) (1, 2)
 - (b) (1, -2)
 - (c) (-1, 2)
 - (d) (-1, -2)
17. Let $A(-2, -1)$ and $B(a, 3)$ be two points, and $|AB| = 5$. Then the value of a is ().
- (a) 1 or -5
 - (b) -1 or 5
 - (c) ± 5
 - (d) ± 1
18. Suppose that a line l passes through two points $M(0, -1)$ and $N(-2, 0)$. Then the equation of l is ().
- (a) $x + y + 2 = 0$
 - (b) $x + 2y + 2 = 0$
 - (c) $2x + y + 2 = 0$
 - (d) $2x + y + 1 = 0$
19. Which of the following inequalities is correct? ().
- (A) $2.1^{\frac{2}{3}} > 1.2^{\frac{2}{3}}$
 - (B) $2.1^{-2} > 1.2^{-2}$
 - (C) $3^{2.25} > 3^3$
 - (D) $0.75^{-0.2} > 0.75^{-0.4}$
20. A circle centered at $A(-3, 2)$, with a radius 3 has equation ().
- (a) $(x + 3)^2 + (y - 2)^2 = 3$
 - (b) $(x + 3)^2 + (y - 2)^2 = 9$
 - (c) $(x - 3)^2 + (y + 2)^2 = 9$
 - (d) $(x - 3)^2 + (y + 2)^2 = 3$

21. Which of the following is correct?

(a) $\tan x = \frac{\sin x}{\cos x} (\cos x \neq 0)$

(b) $\tan x = \frac{1}{\cos x} (\cos x \neq 0)$

(c) $\tan x = \frac{\cos x}{\sin x} (\sin x \neq 0)$

(d) $\tan x = \frac{1}{\sin x} (\sin x \neq 0)$

22. In an arithmetic sequence $\{a_n\}$, if $a_1 = 1$ and its common difference $d = 2$, then $a_{10} = ()$.

(a) 15

(b) 19

(c) 21

(d) 17

23. Let $a > b$. Then which of the following is correct? $()$.

(a) $|a| < |b|$

(b) $|a| > |b|$

(c) $a^3 > b^3$

(d) $a^2 > b^2$

24. A circle has center at $(-3, 2)$ and radius 4. Then the equation of the circle is $()$.

(a) $(x + 3)^2 + (y - 2)^2 = 16$

(b) $(x - 3)^2 + (y + 2)^2 = 16$

(c) $(x - 3)^2 + (y + 2)^2 = 4$

(d) $(x + 3)^2 + (y - 2)^2 = 4$

25. If an exponential function $y = a^x$ is monotonic increasing on $\mathbb{R}$, then the range of a is $()$.

(a) $a < 0$ or $a > 1$

(b) $a > 1$

(c) $0 < a < 1$

(d) $a > 0$

26. If $\sin \alpha = \frac{\sqrt{5}}{3}$, then $\cos 2\alpha = ()$.

(a) $-\frac{2}{9}$

(b) $\frac{1}{9}$

(c) $-\frac{1}{9}$

(d) $\frac{2}{9}$

27. If three lines $ax + 2y + 8 = 0$, $4x + 3y = 10$ and $2x - y = 10$ intersect at one point, then $a = ()$.

(a) 1

(b) -3

(c) 3

(d) -1

28. Let $mx^2 + ny^2 = 1$ be a hyperbola. Then m, n satisfy $()$.

(a) $mn < 1$

(b) $mn > 1$

(c) $mn > 0$

(d) $mn < 0$

29. Let $a = \lg 4$, $b = \lg 5$. Then $a + 2b = ()$.

(a) 100

(b) 4

(c) 2

(d) 5

30. The distance between the points $P(-1, 2)$ and $Q(3, 1)$ is $()$.

(a) $\sqrt{5}$

(b) 5

(c) $\sqrt{13}$

(d) $\sqrt{17}$

31. The minimum positive period of the function $f(x) = 3\cos(4x - \frac{\pi}{3})$ is ().

- (a) $\frac{\pi}{3}$
- (b) $\frac{\pi}{2}$
- (c) $\frac{\pi}{4}$
- (d) 2π

32. The equation of the directrix of the parabola $y = -\frac{1}{4}x^2$ is ().

- (a) $y = 1$
- (b) $y = \frac{1}{16}$
- (c) $x = 1$
- (d) $x = \frac{1}{16}$

33. The line that is parallel to $4x - 2y - 1 = 0$ is ().

- (a) $x + 2y - 1 = 0$
- (b) $2x + y - 1 = 0$
- (c) $x - 2y - 1 = 0$
- (d) $2x - y - 1 = 0$

34. Suppose that $\cos \alpha = -\frac{1}{2}, \alpha \in (\frac{\pi}{2}, \pi)$, then the value of $\sin \frac{\alpha}{2}$ is ().

- (a) $-\frac{1}{2}$
- (b) $\frac{\sqrt{3}}{2}$
- (c) $-\frac{\sqrt{3}}{2}$
- (d) $\frac{1}{2}$

35. Let α, β be acute angles. Then which of the following is correct? ().

- (a) $\sin(\alpha - \beta) < \sin(\alpha + \beta)$
- (b) $\cos(\alpha - \beta) < \cos(\alpha + \beta)$
- (c) $\cos \alpha \cos \beta < \cos(\alpha + \beta)$
- (d) $\sin \alpha + \sin \beta < \sin(\alpha + \beta)$

36. The range of the function $f(x) = \frac{1}{1+|x|}$ ($x \in \mathbb{R}$) is ().

- (a) $(0, 1]$
- (b) $[0, 1]$
- (c) $[0, 1)$
- (d) $(0, 1)$

37. The equation of the directrix of the parabola $y^2 = -x$ is ().

- (a) $x = \frac{1}{4}$
- (b) $x = \frac{1}{2}$
- (c) $x = -\frac{1}{4}$
- (d) $x = -\frac{1}{2}$

38. If $\cos \alpha = \frac{1}{3}$ and $-\frac{\pi}{2} < \alpha < 0$, then $\tan \frac{\alpha}{2} = ()$.

- (a) $\frac{\sqrt{2}}{2}$
- (b) $-\sqrt{2}$
- (c) $-\frac{\sqrt{2}}{2}$
- (d) $\sqrt{2}$

39. If $\sin \alpha = \frac{1}{3}$, then $\cos(\frac{\pi}{2} + \alpha) = ()$.

- (a) $-\frac{2\sqrt{2}}{3}$
- (b) $\frac{2\sqrt{2}}{3}$
- (c) $\frac{1}{3}$
- (d) $-\frac{1}{3}$

40. Which of the following descriptions about the function $f(x) = \tan x$ is incorrect? ().

- (a) $f(x)$ satisfies $f(-x) = -f(x)$ on its domain
- (b) $f(x)$ has exactly one zero on the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$
- (c) $f(x)$ is monotonic increasing over $(-\frac{\pi}{2}, \frac{\pi}{2})$
- (d) $f(x)$ is even

41. In a sequence $\{a_n\}$, if $a_1 = \frac{1}{2}$ and $\frac{1}{a_{n+1}} = \frac{1}{a_n} + 1, n \geq 1$, then $a_6 = ()$.

- (a) $\frac{1}{7}$
- (b) $\frac{1}{8}$
- (c) $\frac{1}{6}$
- (d) $\frac{1}{9}$

42. Let $f(x) = \log_5(2x + 1)$. Then the solution set of $f(x) \geq 0$ is ().

- (a) $\{x|x \geq 1\}$
- (b) $\{x|x > 0\}$
- (c) $\{x|x > 1\}$
- (d) $\{x|x \geq 0\}$

43. Suppose that the major axis of an ellipse is 10, the two foci F_1, F_2 are located at $(3, 0)$ and $(-3, 0)$. The equation of the ellipse is ().

- (a) $\frac{x^2}{9} + \frac{y^2}{25} = 1$
- (b) $\frac{x^2}{25} + \frac{y^2}{16} = 1$
- (c) $\frac{x^2}{16} + \frac{y^2}{25} = 1$
- (d) $\frac{x^2}{16} + \frac{y^2}{9} = 1$

44. Let $\vec{a} = (-2, 3)$, $\vec{b} = (2, 1)$. Then $\vec{a} - 2\vec{b} = ()$.

- (a) $(-6, -1)$
- (b) $(6, -1)$
- (c) $(-6, 1)$
- (d) $(6, 1)$

45. Suppose that a straight line l is perpendicular to another line $l_1 : x + 2y + 4 = 0$, and passes through the intersection of $l_2 : x + y + 1 = 0$ and $l_3 : 2x + y - 1 = 0$. The equation of l is $()$.

- (a) $2x - y - 7 = 0$
- (b) $x + 2y - 3 = 0$
- (c) $x - 2y - 1 = 0$
- (d) $2x + y - 5 = 0$

46. Suppose that z_1 and z_2 are two roots of the equation $z^2 - 2z + 2 = 0$. Let $a_n = (z_1^n - z_2^n)^2$. Then $a_4 = ()$.

- (a) 0
- (b) 32
- (c) -64
- (d) -32

47. Suppose that $a_n = 2^{n+1} - 3n$, $n \geq 1$. Let S_n be the sum of the first n terms of the sequence. Then $S_n = ()$.

- (a) $2^{n+2} - \frac{3n(n+1)}{2} + 4$
- (b) $2^{n+2} - \frac{3n(n+1)}{2} - 4$
- (c) $2^{n+2} - 3n - 4$
- (d) $2^{n+2} + \frac{3n(n+1)}{2} - 4$

48. A bag contains five balls identical in size and shape, including two red balls, labeled 1 and 2; two black balls, labeled 1 and 2; one white ball, labeled 1. Two balls are drawn randomly from the bag. The probability that the balls have different colors, and the probability that at least one of the two balls is labeled 1 are () respectively.

(a) $\frac{4}{5}, \frac{7}{10}$

(b) $\frac{3}{5}, \frac{9}{10}$

(c) $\frac{3}{5}, \frac{9}{10}$

(d) $\frac{4}{5}, \frac{9}{10}$